Xpath cheatsheet

Xpath test bed

Test queries in the Xpath test bed:

Xpath test bed
(whitebeam.org)

Browser console

```
$x("//div")
```

Works in Firefox and Chrome.

Selectors

Descendant selectors

`h1`	`//h1`	?
`div p`	`//div//p`	?
`ul > li`	`//ul/li`	?
`ul > li > a`	`//ul/li/a`	
`div > *`	`//div/*`	
`:root`	`/`	?
`:root > body`	`/body`	

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Attribute selectors

`#id`	`//*[@id="id"]`	?
`.class`	`//*[@class="class"]` …kinda	
`input[type="submit"]`	`//input[@type="submit"]`	
`a#abc[for="xyz"]`	`//a[@id="abc"][@for="xyz"]`	?
`a[rel]`	`//a[@rel]`	
`a[href^='/']`	`//a[starts-with(@href, '/')]`	?
`a[href$='pdf']`	`//a[ends-with(@href, '.pdf')]`	
`a[href*='://']`	`//a[contains(@href, '://')]`	
`a[rel~='help']`	`//a[contains(@rel, 'help')]` …kinda	

Order selectors

`ul > li:first-of-type`	`//ul/li[1]`	?
`ul > li:nth-of-type(2)`	`//ul/li[2]`	
`ul > li:last-of-type`	`//ul/li[last()]`	
`li#id:first-of-type`	`//li[1][@id="id"]`	?
`a:first-child`	`//*[1][name()="a"]`	
`a:last-child`	`//*[last()][name()="a"]`	

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Siblings

`h1 ~ ul`	`//h1/following-sibling::ul`	?
`h1 + ul`	`//h1/following-sibling::ul[1]`	
`h1 ~ #id`	`//h1/following-sibling::[@id="id"]`	

jQuery

`$('ul > li').parent()`	`//ul/li/..`	?
`$('li').closest('section')`	`//li/ancestor-or-self::section`	
`$('a').attr('href')`	`//a/@href`	?
`$('span').text()`	`//span/text()`	

Other things

`h1:not([id])`	`//h1[not(@id)]`	?
Text match	`//button[text()="Submit"]`	?
Text match (substring)	`//button[contains(text(),"Go")]`	
Arithmetic	`//product[@price > 2.50]`	
Has children	`//ul[*]`	
Has children (specific)	`//ul[li]`	
Or logic	`//a[@name or @href]`	?
Union (joins results)	`//a | //div`	?

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Class check

```
//div[contains(concat(' ',normalize-space(@class),' '),' foobar ')]
```

Xpath doesn't have the "check if part of space-separated list" operator, so this is the workaround (source).

Expressions

Steps and axes

`//`	`ul`	`/`	`a[@id='link']`
Axis	Step	Axis	Step

Prefixes

Prefix	Example	What
`//`	`//hr[@class='edge']`	Anywhere
`./`	`./a`	Relative
`/`	`/html/body/div`	Root

Begin your expression with any of these.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Axes

Axis	Example	What
`/`	`//ul/li/a`	Child
`//`	`//[@id="list"]//a`	Descendant

Separate your steps with `/`. Use two (`//`) if you don't want to select direct children.

Steps

```
//div
//div[@name='box']
//[@id='link']
```

A step may have an element name (`div`) and predicates (`[...]`). Both are optional. They can also be these other things:

```
//a/text()      #=> "Go home"
//a/@href       #=> "index.html"
//a/*           #=> All a's child elements
```

Predicates

Predicates

```
//div[true()]
//div[@class="head"]
//div[@class="head"][@id="top"]
```

Restricts a nodeset only if some condition is true. They can be chained.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Operators

```
# Comparison
//a[@id = "xyz"]
//a[@id != "xyz"]
//a[@price > 25]
```

```
# Logic (and/or)
//div[@id="head" and position()=2]
//div[(x and y) or not(z)]
```

Use comparison and logic operators to make conditionals.

Using nodes

```
# Use them inside functions
//ul[count(li) > 2]
//ul[count(li[@class='hide']) > 0]
```

```
# This returns `<ul>` that has a `<li>` child
//ul[li]
```

You can use nodes inside predicates.

Indexing

```
//a[1]                  # first <a>
//a[last()]             # last <a>
//ol/li[2]              # second <li>
//ol/li[position()=2]   # same as above
//ol/li[position()>1]   # :not(:first-of-type)
```

Use `[]` with a number, or `last()` or `position()`.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Chaining order

```
a[1][@href='/']
a[@href='/'][1]
```

Order is significant, these two are different.

Nesting predicates

```
//section[.//h1[@id='hi']]
```

This returns `<section>` if it has an `<h1>` descendant with `id='hi'`.

Functions

Node functions

```
name()                  # //[starts-with(name(), 'h')]
text()                  # //button[text()="Submit"]
                        # //button/text()
lang(str)
namespace-uri()
```

```
count()                 # //table[count(tr)=1]
position()              # //ol/li[position()=2]
```

Boolean functions

```
not(expr)               # button[not(starts-with(text(),"Submit"))]
```

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

String functions

```
contains()                # font[contains(@class,"head")]
starts-with()             # font[starts-with(@class,"head")]
ends-with()               # font[ends-with(@class,"head")]
```

```
concat(x,y)
substring(str, start, len)
substring-before("01/02", "/")  #=> 01
substring-after("01/02", "/")   #=> 02
translate()
normalize-space()
string-length()
```

Type conversion

```
string()
number()
boolean()
```

Axes

Using axes

```
//ul/li                     # ul > li
//ul/child::li              # ul > li (same)
//ul/following-sibling::li  # ul ~ li
//ul/descendant-or-self::li # ul li
//ul/ancestor-or-self::li   # $('ul').closest('li')
```

Steps of an expression are separated by `/`, usually used to pick child nodes. That's not always true: you can specify a different "axis" with `::`.

`//`	`ul`	`/child::`	`li`
Axis	Step	Axis	Step

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Child axis

```
# both the same
//ul/li/a
//child::ul/child::li/child::a
```

`child::` is the default axis. This makes `//a/b/c` work.

```
# both the same
# this works because `child::li` is truthy, so the predicate succeeds
//ul[li]
//ul[child::li]
```

```
# both the same
//ul[count(li) > 2]
//ul[count(child::li) > 2]
```

Descendant-or-self axis

```
# both the same
//div//h4
//div/descendant-or-self::h4
```

`//` is short for the `descendant-or-self::` axis.

```
# both the same
//ul//[last()]
//ul/descendant-or-self::[last()]
```

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Other axes

Axis	Abbrev	Notes
`ancestor`		
`ancestor-or-self`		
`attribute`	`@`	`@href` is short for `attribute::href`
`child`		`div` is short for `child::div`
`descendant`		
`descendant-or-self`	`//`	`//` is short for `/descendant-or-self::node()/`
`namespace`		
`self`	`.`	`.` is short for `self::node()`
`parent`	`..`	`..` is short for `parent::node()`
`following`		
`following-sibling`		
`preceding`		
`preceding-sibling`		

There are other axes you can use.

Unions

```
//a | //span
```

Use | to join two expressions.

More examples

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Examples

```
//*                 # all elements
count(//*)          # count all elements
(//h1)[1]/text()    # text of the first h1 heading
//li[span]          # find a <li> with an <span> inside it
                    # ...expands to //li[child::span]
//ul/li/..          # use .. to select a parent
```

Find a parent

```
//section[h1[@id='section-name']]
```

Finds a `<section>` that directly contains `h1#section-name`

```
//section[//h1[@id='section-name']]
```

Finds a `<section>` that contains `h1#section-name`. (Same as above, but uses descendant-or-self instead of child)

Closest

```
./ancestor-or-self::[@class="box"]
```

Works like jQuery's `$().closest('.box')`.

Attributes

```
//item[@price > 2*@discount]
```

Finds `<item>` and check its attributes

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

References

- Xpath test bed (whitebeam.org)

▶ **21 Comments** for this cheatsheet. Write yours!

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd **HTML to PDF**